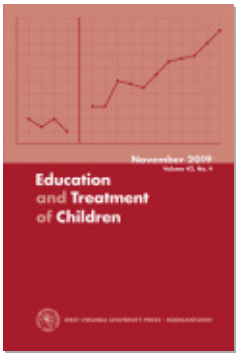
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Video Modeling Using Classroom Peers as Models to Increase Social Communication Skills in Children with ASD in an Integrated Preschool

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Abstract

This study examined if a functional relationship exists between video modeling (VM) of peer models and increased outcomes in social communication goals for preschool-age children in an integrated public school setting. Six preschool students received the VM treatment in the classroom using a withdrawal design (A-B-A-B-C); two additional students (comparison group) received treatment-as-usual (i.e., group instruction with individual support). All six treatment group students demonstrated an increasing trend across treatment sessions, with higher response rates than during baseline sessions; two participants showed less variability and a positive trend in the pull-out setting with fewer distractions. The two comparison participants did not show any gains from treatment-as-usual; however, positive gains were immediately noted after subsequently introducing VM. Using peer models in VM increased attention to videos and generalized to imitation of other social communication behaviors. Both push-in (i.e., individual treatment in classrooms) and pull-out sessions proved beneficial.

Keywords: video modeling, peer mediated instruction, preschool, social communication

At the turn of the 21st century, only 1 out of 150 people was being diagnosed with autism spectrum disorder (ASD; Centers for Disease Control and Prevention [CDC], 2017). Today, ASD is one of

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the fastest growing developmental disabilities in the United States, and the rate of diagnosis has almost tripled, with 1 in 68 children currently being diagnosed (CDC, 2017). According to the *Diagnostic and Statistical Manual of Mental Disorders—Fifth Edition* (American Psychiatric Association, 2013), an individual diagnosed with ASD displays persistent deficits in social communication and social interaction and exhibits restricted, repetitive patterns of behaviors or interests. As evidence-based practices have been designed to improve the outcomes for children with disabilities, the Council for Exceptional Children (CEC, 2014) created standards for classifying evidence-based practices in special education. Regarding autism, the National Autism Standards (National Autism Center, 2015) and the National Professional Development Center on Autism Spectrum Disorder (NPDC, 2011) completed systematic reviews and identified evidence-based practices to teach children with ASD foundational skills. The NPDC employed the research design quality indicators established by CEC in 2014 (Wong et al., 2013). Video modeling (VM) and peer-mediated interventions (PMI) are among the evidence-based practices shown to be very effective when teaching students with ASD across a variety of skills (NPDC, 2011).

Video Modeling

VM has been implemented and presented in the literature for almost 20 years and has demonstrated success in helping children with ASD learn social communication skills (Bellini & Akullian, 2007; Cardon, 2012, 2013; Cardon & Wilcox, 2011; Charlop-Christy, Le, & Freeman, 2000; MacDonald, Sacramone, Mansfield, Wiltz, & Ahearn, 2009; Nikopoulos & Keenan, 2004). VM is supported by Bandura's (1977) social learning theory. Modeling is defined as the process of a model—live, recorded, and/or imagined—demonstrating the behaviors that the learner will replicate (Corbett & Abdullah, 2005; Delano, 2007). During VM, a student with ASD watches a video of the target skill and is then asked to demonstrate the target behavior (Charlop-Christy et al., 2000). According to Bellini and Akullian's (2007) review, children with ASD respond well to peers or adults who serve as the video model and can learn a variety of skills, including imitation, self-help skills, communication skills, play skills, and social skills. Some research has indicated that children with autism learn more quickly and generalize more via VM (Cardon & Wilcox, 2011; Charlop-Christy et al., 2000), yet other research has found no difference between direct modeling and video modeling when implemented by peers in a classroom setting (Odluyurt, 2013).

Peer-Mediated Intervention

In PMI, neurotypical peers are trained to model various skills, such as social interactions or responding to social initiations (Chan et al., 2009). In 2009, Chan et al. reviewed 42 studies in which PMI was implemented with 172 target students with ASD. Overall, 91% of the target students improved social skill behaviors. Furthermore, Change and Locke's (2016) systematic review of PMI for children with ASD recognized PMI as a very successful intervention methodology for improving social skills. Harjusola-Webb, Parke Hubbell, and Bedesem (2012) stated that "using peers as role models has the potential to be more advantageous than teacher modeling for increasing the quality and quantity of social behaviors in natural environments" (p. 30).

Video Modeling and Peer-Mediated Intervention

Despite the growing body of literature supporting VM, with peers acting as the video model, in conjunction with PMI (Kourassanis, Jones, & Fienup, 2014; Maione & Mirenda, 2006; Marcus & Wilder, 2009; Nikopoulos & Keenan, 2004), very few studies have used VM in conjunction with PMI using peers from an integrated or inclusive preschool setting (e.g., Odluyurt, 2013). Marcus and Wilder (2009) found that the use of peer models enhanced performance, and Kourassanis et al. (2014) found that peer VM increased the attention of children with ASD when watching videos as well as their rate of acquisition and/or improved generalization. New research is needed to analyze integrated preschool educational environments with peers as video models as there is often a push to place students with autism with their non-disabled peers as an appropriate least-restrictive environment.

Integrated Educational Environments

The Individuals with Disabilities Act (IDEA) states that students must be educated alongside non-disabled peers in regular education environments to the greatest extent possible (IDEA, 2004; Yell, 1995); however, it is not clear how, or if all children with autism, benefit from full-time integrated classrooms. Research analyzing evidence-based practices in inclusive preschool settings has demonstrated that students' needs can be met alongside their peers. For example, Stahmer, Akshoomoff, and Cunningham (2011) examined the effectiveness of an inclusive toddler program utilizing pivotal response training and picture exchange communication systems for 102 children diagnosed with ASD or a pervasive developmental disorder not otherwise

specified (PDD-NOS). Standardized rating scales were used to monitor the children's progress. By the time participants exited the program, 31% of the participants were functioning in a typical range. By the end of the program, the participants averaged 16% above the expected rate of development for children with ASD. The researchers concluded that the study "provides support for a community inclusion model that combines evidence-based intervention methods and if children with ASD benefit from inclusion in the preschool years, inclusion in the toddler years may also increase social and language behaviors" (Stahmer et al., 2011, p. 637). What is not clear is how treatment was adjusted to accommodate learners with ASD who continued to struggle in the integrated environment. Research identifying positive outcomes for students with ASD in inclusive settings is limited (Ferraioli & Harris, 2011), although several published articles have identified how integrative settings can support learners with ASD (Leach & Duffy, 2009; Simpson, de Boer-Ott, & Smith-Myles, 2003). The current research analyzed a combination of push-in (i.e., individual treatment in the classroom setting) and pull-out sessions in a preschool environment to accommodate individual learners' needs.

This research combined expansive research on VM in an integrated preschool setting utilizing peer models as the video models. Both push-in and pull-out options were utilized to determine if accommodations for the integrated setting were required. Specifically, the goal of this study was to determine if a functional relationship existed between VM of peer models and increased outcomes on social communication goals for preschool-age children in an integrated public school setting.

Method

Participants

Participants with ASD originally included seven boys and one girl; their ages ranged from 39 to 52 months old (see Table 1). All participants were attending an integrated classroom that was part of a local school district but located on a university campus, and were a sample of convenience given their classroom location on a university campus. The integrated classroom had two sessions: one Early Bird morning group and one Later Gator afternoon group. The classroom, teacher, and aides were consistent across both groups. To be placed in the integrated autism classroom, participants were tested and observed

Table 1
Participants' Demographic Data

Participant	Gender	Age (in years)	Diagnosis	Group
Kevin	Male	4	ASD	Later Gator
Ricky	Male	4	ASD	Later Gator
Andy	Male	3	ASD	Later Gator
Adrian	Male	3	ASD	Later Gator
Sue	Female	4	ASD	Later Gator
Wes	Male	4	ASD	Later Gator
Kade	Male	3	ASD	Early Bird
Hank	Male	4	ASD	Early Bird

Note. ASD = autism spectrum disorder.

by the local school district team and received an educational classification of ASD. The district administered the Gilliam Autism Rating Scale-3 (GARS-3; Gilliam 2014) and the Behavior Assessment for Children-3 (BASC-3; Reynolds & Kamphaus, 2015). Children had to place in the "very likely" range on the GARS-3 and the "clinically significant" range on the BASC-3 for inclusion in the integrated autism preschool. The children were included in the study if they (a) were placed in an integrated autism preschool classroom based on a district classification of ASD and (b) had social communication goals indicated by their district's individual education plans (IEPs). Participants (n = 2) in the Early Bird classroom acted as a comparison group and received treatment-as-usual (i.e., group instruction with individual support during center time) in their classroom. The Early Birds did not receive the VM treatment until after the Later Gator group completed treatment and only after it was determined that they had not made progress with the treatment-as-usual method. The Later Gators (n = 6) received the VM treatment using a withdrawal design (described subsequently). The peers who participated as the video models attended the Later Gator session. To protect participants' confidentiality, pseudonyms have been used throughout this paper.

Peers. In this study, neurotypical classroom peers were trained to be the models in the videos. The peers were recorded demonstrating the target skills to be learned by the preschoolers with ASD. The videos were then shown to the students with ASD, who could watch consistent visual demonstrations of the desired skill multiple times until they could model or imitate the target behavior.

Table 2
Target Behaviors

Target Skill	Sd	Response (within 10 s)
Sharing Puzzle Pieces	Place empty puzzle board in front of participant.	Child verbally requests (approximations or target response) access to a puzzle piece.
Cooperative Block Building	Clinician states, "Let's play," while placing blocks in front of child and placing an initial block for building together.	Child places a second block on the initial block without taking the block off or moving away to play alone.
Transition	Timer beeps; educator states, "Time to clean up," followed by "Time to line up."	Child lines up on the tape near the center and waits to be handed a transition card.

Note. The lead teacher identified the above target skills for all eight participants.
Sd = discriminative stimulus.

Settings and Materials

All baseline and primary treatment sessions were conducted in a preschool classroom on a university campus. The classroom had three areas (centers) with tables and chairs surrounded by bookshelves and one center established by a rug on the floor. The preschool included a teacher desk, cabinets, and cubbies. In addition, a bathroom and an observation room were attached to the classroom. Participants completed the tasks on the rug or at a table, depending on where their "center rotation" occurred. The classroom was identified as an integrated classroom in which children with ASD and neurotypical peers were also enrolled. The morning session (Early Birds) included two children with ASD and two neurotypical peers for part of the day. The afternoon session (Later Gators) included eight children: six identified with ASD and two neurotypical peers. The classroom had one lead teacher, a lead paraprofessional, and two student interns.

Target behaviors (see Table 2) were selected by the preschool teacher and the researchers during a review of the participants' IEP goals. Materials for the target behaviors were wood blocks, classroom puzzles, and a timer with a sound to indicate transition times. A piece of tape was placed on the floor to visually indicate where participants were to line up during transition times. An iPad (version 9.7) and an HDE Shock Proof iPad Case for Kids Bumper Cover with Handle Stands were used to record and display the video models. Standard video recording software on the iPad was used to record the videos.

Video models included two typical peers in the integrated classroom. The three video clips were 18, 19, and 34 s long. Identical clips of the video models were airdropped onto a total of four iPads so that each research assistant (RA) had access to them.

Procedures

Training. Four student RAs (three undergraduate, one graduate) attended a 1.5-hr training session to learn about video modeling, creating video models, and video modeling implementation. They also received a training manual with instructions. The RAs reviewed the protocols for the baseline and treatment. The primary researchers consulted with the classroom teacher to identify appropriate target behaviors given the social communication goals on the participants' IEPs.

Creation of the video models. Two typical peers were taught a brief script (e.g., "let's play," "can I have a piece") to coincide with their actions. The peers practiced the target skills several times, with coaching provided by the research team. The peers were then both recorded interacting with each other to demonstrate the social communication skills being targeted.

Baseline (A). Five baseline sessions were conducted with all eight participants. Baseline sessions occurred consecutively over five days when the child was in attendance. Due to absences that occurred at the beginning of the study, several children extended their baseline sessions into the following week. During the first baseline session, the RAs received live coaching and feedback. During the baseline assessment, all of the target behaviors were randomly presented for the children to demonstrate. An RA would deliver the discriminative stimulus (SD; e.g., "Let's play" while stacking a block) and wait 10 s for the participant to respond. If the child responded with an appropriate action, defined as copying the target action with an action that looked distinctly like the action being modeled, within 10 s and before they performed a different action, then the child was told "thanks," and the RA moved on to the next behavior (Cardon, 2012, 2013; Cardon & Wilcox, 2011; Ingersoll & Schreibman, 2006). If the child did not respond, then they were told "thanks," and the RA moved on to the next behavior. In total, 10 baseline sessions, five consecutive sessions across two different time points, were completed for six Later Gators and two Early Birds.

Treatment (B). Five treatment sessions were conducted once initial baseline sessions were completed with the six Later Gator participants. The two Early Birds in the study did not receive VM treatment at this time. Instead, they received treatment-as-usual in

their preschool classroom. Treatment-as-usual consisted of "center time," where social communication skills were addressed during three to six rotations per day. Each participant also received push-in (i.e., individual treatment in the classroom setting) speech and language services for 30 min per week. Treatment sessions for the Later Gators took place during the hour-long afternoon center rotation time in the classroom setting. During the first treatment session, the RAs received live coaching and feedback from the study authors during implementation. The RA would show a recorded clip of the typical peer performing the target behavior, deliver the SD, and wait 10 s for the participant to respond. The participant could view the video up to three times if the target response was not elicited. Target behaviors consisted of sharing (puzzles), engaging in cooperative play (wood blocks), and transitioning between centers (timer and tape on floor). Video clips were alternated randomly throughout treatment to account for any possible order effects. All participants were presented with all three target behaviors.

Return to baseline (A). All six Later Gators returned to baseline after the initial treatment. The return to baseline sessions were identical to the initial baseline sessions with no iPads or VMs present during each task, for a total of 10 baseline sessions. Baseline sessions on the two remaining Early Birds were also conducted during this time to assess how treatment-as-usual impacted their ability to perform the target behaviors.

Treatment (B). After the return to baseline, all six Later Gators returned to VM treatment that was identical to the initial treatment sessions. The sessions took place in the classroom during center rotation time, and the exact same peer-modeled videos were utilized to target the identified behaviors. Early Bird participants received treatment-as-usual during this phase.

Treatment pull-out (C). After a traditional A-B-A-B design, a third individual treatment session (C) was implemented. During this treatment phase, the same video clips of peers were utilized; however, treatment took place one-on-one with a clinician in a pull-out treatment room next to the preschool classroom. The treatment room had a child-sized table and chairs and was approximately 10 ft by 10 ft in size. Due to the nature of the pull-out setting, only two behaviors (sharing and cooperative play) were targeted during these treatment sessions. Again, Early Bird participants received treatment-as-usual during this phase.

Maintenance and generalization. After the completion of the two baseline phases and all three treatment phases, follow-up sessions were initiated after four and eight weeks. During the follow-up

Table 3
Generalization Behaviors

Target Skill	Sd	Response (within 10 s)
Sharing Puzzle Pieces	Place blank piece of paper in front of the child.	Child verbally requests (approximations or target response) to get access to a marker.
Cooperative Block Building	Clinician states, "Let's play," while placing large Legos in front of child and placing an initial Lego for building together.	Child places a second Lego on the initial Lego without taking the Lego off or moving away to play alone.
Transition	Bell rings on playground; educator states, "Time to line up."	Child lines up on the tape near the door and waits to be handed a transition card.

Note. Sd = discriminative stimulus.

sessions, both the maintenance of the targeted skills and the generalization to novel exemplars were explored. During maintenance, the target skills (sharing, cooperative play, and transition) were reintroduced without the VM in the classroom and pull-out setting. The target SD was delivered, and participants' responses were recorded. If participants failed to respond without the VM, then the VM was reintroduced to see if skills were maintained with the VM as a prompt. For generalization purposes, novel exemplars, determined by the primary investigators and the preschool teacher to be similar in nature, were introduced (see Table 3). During the generalization phase, the novel exemplars were introduced without a VM, and participants' responses were recorded.

Early Bird sessions. The two Early Bird participants who acted as controls were included in two baseline phases (for a total of 10 baseline sessions) that occurred weeks apart. During the weeks when the Later Gators were receiving VM treatment, the Early Bird participants received treatment-as-usual in their classroom. Once Treatment C was completed with the Later Gators, the Early Bird participants received two treatment phases (for a total of 10 treatment sessions) of VM treatment.

Intensity/duration. Participants were scheduled to be seen once per day, five days per week, during the baseline and treatment phases. On average, it took less than 10 min per participant, per session, for all three target behaviors to be addressed. Although some participant absences and holidays prohibited strict adherence to the proposed

Table 4
Fidelity of Treatment

Category	Percent Correct
Play the video clip	100%
Administer the Sd	100%
Wait 10 s	100%
Correct/Incorrect	99.3%
Verbal praise	100%
Physical prompt	99.9%
Verbal praise after physical prompt	99.9%

Note. Sd = discriminative stimulus.

five-days-per-week schedule, all of the Later Gator participants completed two baseline and three treatment phases of the study in just under two months.

Fidelity and reliability. To ensure fidelity of implementation by the RAs, point-by-point comparisons were calculated across participants and sessions. Of the 120 baseline sessions, 30% (k = 36) were scored by the primary researchers live in the classroom setting. Reliability standards greater than 80% were easily met during baseline, with actions being modeled correctly by RAs 100% of the time, waiting 10 s 100% of the time, and imitation rated as correct/incorrect 99.4% of the time. Fidelity across the 90 treatment sessions was calculated for 47% of sessions (k = 43). The primary researchers conducted fidelity checks across all three treatment phases. The fidelity of treatment can be viewed in Table 4, with all categories greater than 95%.

Social validity. To document the social validity of the VM treatment involving peers in the integrated classroom, the preschool teacher and classroom aides provided a qualitative analysis of instances of peer imitation that were observed after VM treatment had been implemented. They were instructed to document specific, novel instances in which a participant with autism imitated the actions of one of the peers filmed in the VM.

Experimental Design and Analysis

A withdrawal design (A-B-A-B-C), with a final session in which participants received treatment in a separate room, was selected for the study. Although social communication skills can be learned and

maintained over time, indicating a possible preclusion from a complete return to baseline, given the addition of a comparison group (the Early Bird preschool class) and the ability to visually analyze the trend, level, and variability between baseline and treatment phases to determine changes and maintained abilities in the target behaviors over time, a withdrawal design was deemed appropriate (Kratochwill et al., 2010). The improvement rate difference (IRD; Parker, Vannest, & Brown, 2009) was calculated as a measure of effect size from nonoverlapping data for analyzing single subject research.

Results

Visual Analysis of Withdrawal Design

ABAB design. A visual analysis of each participant's performance was conducted. All six Later Gators demonstrated an increasing trend across treatment sessions with higher rates of response than what occurred during baseline sessions. Two of the participants (Ricky and Sue) showed response levels greater than 50% in the pull-out setting (Treatment C), when fewer distractions were present in comparison to the classroom setting. The two Early Bird participants did not show any gains of target behaviors when engaged in treatment-as-usual; however, positive gains were noticed immediately after the VM was introduced. A complete return to original baseline levels was not expected during the second baseline phase as it was possible that some social communication skills would be maintained from the first treatment phase. After a visual analysis of the data, a functional relationship between VM and increased social communication skills was noted across all six participants (see Figure 1) with Ricky showing an increase in level after implementation of Treatment C. Increasing trends, limited variability, and immediate response to the VM are evident in the individual graphs.

Treatment C. After visually analyzing the data from the first four weeks of baseline and treatment, it was determined that a third treatment session would be beneficial. All six participants viewed the cooperative play and sharing videos in a pull-out treatment room. Although all six participants responded to the VM in the pull-out treatment sessions, two of the participants (Sue and Ricky) responded more consistently and demonstrated less variability in the distraction-free environment. Adrian demonstrated more variability throughout his sessions, due in part to a suspected diagnosis of childhood apraxia of speech. At times, it was difficult to interpret his correct versus

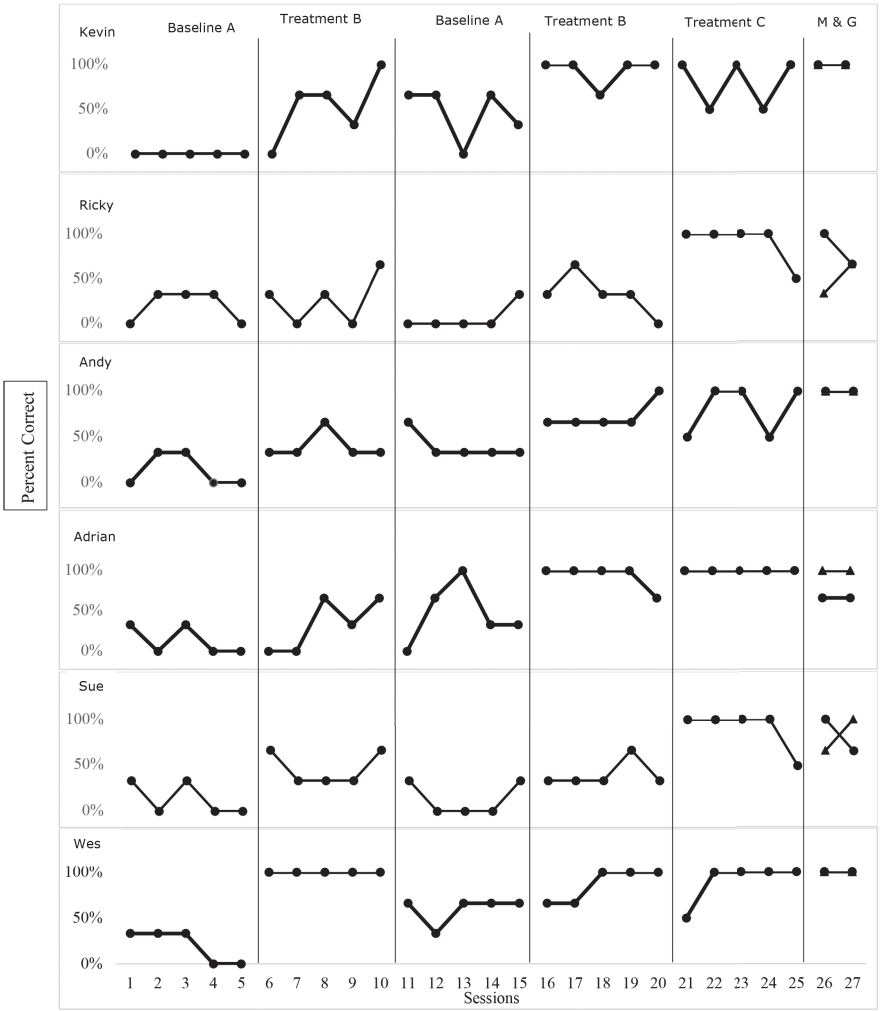


Figure 1. Withdrawal design graph for the Later Gator group.

Note. M = Maintenance represented by a circle; G = Generalization represented by a triangle.

incorrect responses to the puzzle piece request behavior, so caution was used in assigning a correct response.

Early Birds. Baseline measures for the Early Birds indicated no changes in target behaviors over the course of four weeks (Phase A). The VM treatment protocol was implemented after the treatment-as-usual demonstrated no effect (Phase B). A trained RA implemented

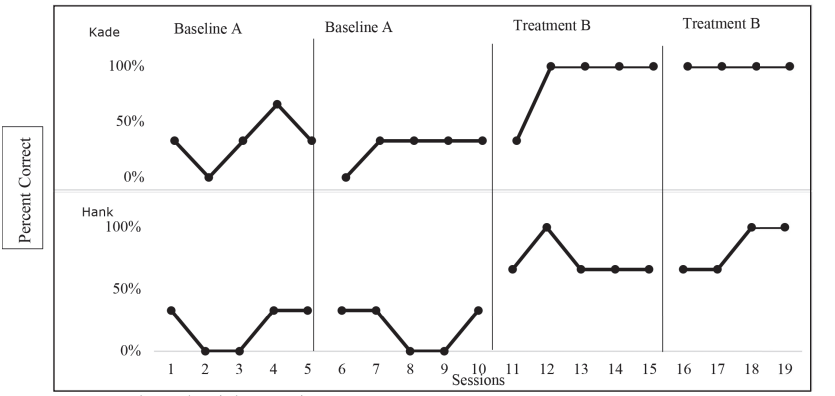


Figure 2. The Early Bird comparison group.

Note. Early Birds did not receive peer-mediated video modeling until after the Later Gators completed the treatment; therefore, their graphs do not represent a traditional withdrawal design.

the VM treatment protocol for the three target behaviors with the two Early Bird students. Early Bird participants made immediate gains in the target behaviors after exposure to the VM intervention indicating an immediacy effect when using VM to support skill acquisition. Baseline measures and results can be seen in Figure 2.

Maintenance and generalization. All six of the participants were able to maintain the target social communication skills at four and six weeks. Use of the VM as a prompt was only required twice during the 18 maintenance opportunities. Generalization to novel exemplars was evident in all six of the participants, with two participants struggling to generalize the transition skill (see Figure 1). VMs were not used in generalization tasks.

IRD. An increased IRD for all six Later Gators was noted during treatment sessions, with lower rates of target skills present during baseline (see Table 5). Only one baseline measure for Ricky was noted to be at equal improvement rate levels. Ricky demonstrated more variability than other participants throughout his sessions and struggled with attention to tasks throughout his day at the preschool.

Social Validity

The preschool teacher and classroom aides were asked to document instances of peer imitation observed after VM treatment was implemented to determine the social validity aspects of using peers as

Table 5
Improvement Rate Difference

Participant	Kevin	Ricky	Andy	Adrian	Sue	Wes	Kade	Hank
Baseline (A_1)	0	60%	40%	60%	60%	60%		
Tx (B_1)	80%	60%	100%	60%	100%	100%	100%	100%
Baseline (A_2)	80%	20%	100%	80%	40%	100%		
Tx (B_2)	100%	80%	100%	100%	100%	100%	100%	100%
Tx (C)	100%	100%	100%	100%	100%	100%		
Maintenance and Generalization	100%	100%	100%	100%	100%	100%		

Note. The improvement rate difference is calculated as number of improved data points in the phase divided by the total number of data points in the phase. A percent improvement greater than zero occurs in each phase.

video models in an integrated classroom. Multiple instances of participants imitating the peer models occurred and are documented in Table 6.

Discussion

This study was designed to examine whether implementing VM with peer mentors would improve the social communication skills of preschool students with ASD. Results of this study indicated that VM with classroom peers as the video models can be an effective and inclusive intervention strategy to teach social communication skills to preschool students with ASD. All participants increased their targeted social communication skills and generalized to social imitation of peers. Participants who received VM treatment demonstrated an increase in targeted social communication skills (engaging in cooperative play, sharing, and following directions) whereas the participants who received treatment-as-usual did not show an increase until after the VM intervention was implemented.

Overall, these findings are consistent with previous research showing that VM can be used in conjunction with peer mentors who act as the models in the videos (Ganz & Flores, 2008; Hine & Wolery, 2006; Jung, Sainato, & Davis, 2008; Katz & Girolametto, 2013;

Table 6
Social Validity—Documentation of Peer Imitation

Behavior	Description
Trick or Treating	Participants imitated the peer models knocking on office doors and holding up their bags.
Hugging	The participant Andy started to imitate the peer models by hugging their mothers.
Face Holding	The participant Andy imitated one of the peers putting her hands on either side of another peer's face.
Playground Activities	Four of the participants followed the peer models around the playground and in PE and imitated a variety of actions.
Attending	One of the participants started to attend to a peer model during center time when emotions cards were being described.
Play	One of the participants started to imitate the peer model's toy play and block building. The participant Adrian imitated the car the peer built and then drove it to the same places where the peer was.
Interaction	Several participants started interacting more consistently with the peer models throughout the day. Adrian and Ricky followed a peer model around the classroom, attempting to give her a dress they thought she wanted. Wes asked a peer model to read a book with him. When the peer declined because he was playing in the kitchen, Wes joined him.
Requesting (Manding)	The participant Wes kept asking the teacher for puzzle pieces, saying, "Can I have a piece?"
Initiating Social Interaction	The participant Sue was playing out on the playground and stopped to give the teacher a hug.
Imitation	The participant Wes watched the peer model to see how he was supposed to be sitting. He then put his feet in the same position as the peer. The participant Adrian followed the peer around to the correct center when it was time to rotate. When the peer model was playing hopscotch, Adrian copied his foot movements.

Note. According to the preschool teacher's report, none of these behaviors were occurring prior to the treatment. PE = physical education.

Kourassanis et al., 2014; Maione & Mirenda, 2006; Marcus & Wilder, 2009; Nikopoulos & Keenan, 2004). These studies demonstrated that using peer models could increase attention to videos and performance. The present study extends this growing body of research because participants imitated what they saw in the VM and then went on to become social observers, as evidenced by the numerous imitation examples documented by the teacher (see Table 6).

These results have clear implications for treatment designed to increase social communication skills in children with ASD. To begin with, limited research has utilized VM or peer VM in an integrated preschool classroom setting. This research indicates that peer-mediated VM (i.e., using classroom peers as the video model) is an effective tool for integrated preschool settings and that treatment options should be individualized to the learner. In addition, although integrative and inclusive settings are important, particularly when looking at the least restrictive environments and special education placements, some learners still require settings that are less distracting, as evidenced by the two participants (Ricky and Sue) who responded more consistently in the pull-out setting. This new information adds to existing research on how to individualize support for learners with ASD in integrated settings.

The current study also indicated that skills learned during peer-mediated VM can be maintained—at least over short periods of time—and can be generalized to new skills. Parents and educators have expressed concerns that children with ASD can regress or lose skills they once had (Meilleur & Fombonne, 2009). In addition, some research indicates that children with ASD learn best via discrete, individual skills taught through daily exposures (e.g., Downs, Downs, Fossum, & Rau, 2008); however, there is an ongoing body of research that indicates children with ASD can be taught pivotal skills that they maintain and generalize to new, previously untargeted skills (Mohammadzaheri, Koegel, Rezaee, & Rafiee, 2014). The evidence that skills learned via VM can generalize to new and untargeted skills is an exciting trend (e.g., Ayres & Langone, 2005; Cardon, 2012; Cardon & Wilcox, 2011). Furthermore, using peer-mediated VM may contribute to the imitation of peers in other instances, as indicated by the social validity measures in this study.

This study was designed to examine the effect of VM with peers on social communication skills. It is important to remember that as part of a single case design, participants acted as their own controls. Although the Early Birds acted as a comparison group, the sample sizes were too small to draw conclusions. Moreover, this was a sample of convenience in a public school in a university campus classroom.

As this study was administered in a public school classroom at the beginning of the school year, students' level of response to the videos over time could have been influenced by getting to know the peers and the classroom routine. The small number of participants limits the extent to which the findings of this study may be generalized. As such, this study needs to be replicated with other children with ASD in similar and/or other settings. Finally, all of the participants in the treatment portion of the study had an educational diagnosis of autism and were placed by the school district in the integrated classroom. The severity levels of their autism diagnosis were not known, which is a limitation of this study.

Future research should focus on peer-mediated VM in integrated settings with elementary age children. In addition, further investigation into VM with peers with whom students already have an established relationship would be beneficial for teasing out any potential confounds. The replication of this study in other integrated public preschool settings is also warranted.

The present study indicates that using a peer-mediated VM can support social communication development in preschool-age children with ASD. Utilizing peers can also contribute to the overall imitation development of children with ASD and give them numerous opportunities to imitate the social interactions they observe with the typical peers in their classroom. Finally, push-in and pull-out settings may be beneficial for children with ASD and should be considered on a case-by-case basis.

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